Calcurion Physics Worksheets

A-Level Physics

Topic: Particles & Radiation

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Name the four fundamental interactions. **[2]**

2. What is a hadron? Give two examples. **[2]**

3. What is a lepton? Give two examples. **[2]**

4. State the quark composition of a proton and a neutron. **[2]**

5. What is an antiparticle? Give an example. **[2]**

6. State the equation for pair production. **[2]**

7. Calculate the minimum energy of a photon for electron–positron pair production. ($m_e = 9.1 \times 10^{-31}$ kg, $c = 3.0 \times 10^{8}$ m/s) **[3]**

8. What is annihilation? Write the equation for electron–positron annihilation. **[2]**

9. Define specific charge. **[1]**

10. Calculate the specific charge of a proton. ($e = 1.6 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg) **[2]**

11. Name the three types of radioactive emission and their composition. **[3]**

12. State the inverse square law for gamma radiation intensity. **[1]**

13. Intensity is 8 units at 1.0 m. What is it at 2.0 m? **[2]**

14. Define the electronvolt. **[1]**

15. Convert 5.0 MeV to joules. **[2]**

Section B – Hard (Q16–25)

16. A π^+ meson has quark composition $u\bar{d}$. What is the composition of a π^-? **[2]**

17. Explain why a free neutron decays but a free proton does not (in terms of mass/energy). **[3]**

18. State the conservation laws that apply in particle interactions. **[3]**

19. In β^- decay a neutron changes to a proton. Write the quark-level equation. **[3]**

20. A photon of energy 2.0 MeV undergoes pair production. Calculate the total kinetic energy of the electron–positron pair. **[3]**

21. Explain the role of the weak interaction in beta decay. **[3]**

22. Calculate the de Broglie wavelength of an electron accelerated through 100 V. ($h = 6.63 \times 10^{-34}$, $m_e = 9.1 \times 10^{-31}$) **[4]**

23. State the photoelectric equation and define each term. **[3]**

24. Light of wavelength 400 nm is incident on a metal with work function 2.0 eV. Calculate the maximum KE of photoelectrons. **[4]**

25. Explain why the existence of antineutrinos was proposed in beta decay. **[3]**

End of Worksheet • *Show all working clearly.*